POAMS: LECTURE 20
MORE ON THE EFFECT OF SPIN ON MOTION

In this lecture, I will continue to present, in Normal Realist terms, the effect of spin on the orbital parameters of free-moving particles. I will present a second numerical example, this time concerned with the effects of spin on a particle located on or near to the surface of Earth. This lecture is based on some of Chapter 7 of 'Light-speed….'.

1. Introduction

Consider, once again, the case of a freely moving particle, P, of mass m, in a closed orbit about a body of mass M, with $m \ll M$. As in the last lecture, we only consider the case in which P is an *axially symmetric* particle with *constant angular speed* in a *circular* orbit about the body B. If P spins (rotates) on its axis while orbiting, then according to POAMS, since any orbital angular momentum is balanced against the total angular momentum of the system, P's spin will cause it take up a natural orbit different from that which it would follow if it were not spinning.

In this lecture, I will present another numerical example, this time involving a possible terrestrial experiment, similar to the one performed by Hideo Hayasaka and his team. Until relatively recently, such experiments were the only ones, as far as we know, which provided a practical way of verifying or disproving the predictions made by POAMS in relation to spin. It is for this reason that the example appears in 'Light-Speed…' and as such, plays a significant part in the history of POAMS.

It is to be emphasised here, as in any of the preceding numerical examples, that there may be inaccuracies in the values of the known physical constants we have used. For example, the currently accepted value of the Newtonian 'gravitational constant', G, is determined accurately by *terrestria*l (*i.e*. constrained) experiments, and as such, may require a minor correction in POAMS. For this reason, the predictions of POAMS should be taken as somewhat more qualitative then quantitative.

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2. Example Two

Our next numerical example concerns the effect of spin on a particle located on or near to the surface of Earth, at the equator. In effect, this example combines the ideas of both Example One of the previous lecture and Example Two of Lecture 16. As in the experiments conducted by Hideo Hayasaka and his Japanese team at Tokai University in 1989, this example is concerned with a *circular disc* of mass *m* kg and of radius $\rho \cong 3.3286$ centimetres, so that $\rho \cong 0.033286$ m. Of course, in general, a disc of any radius may be considered. In this case, the radius of the disc's approximately circular constrained orbit is the Earth's mean equatorial radius, given by $r \cong 6.378137 \times 10^6$ m, and its orbital speed is the rotational speed of Earth, namely $v \cong 465.111111$ m s^{-1}. It follows by equation (7) of Lecture 16 [$L = mvr$], that the magnitude of the disc's orbital angular momentum in this constrained orbit is

$$L = mvr \cong 2.966542386m \times 10^9 \text{ kg m}^2 \text{ s}^{-1}$$

Let us suppose, for the moment, that the disc is an inert mass which does not spin. If follows by equation (9) of Lecture 16 [$G' = rv^2/M$], exactly as with the 1 kg mass of Example Two, that the 'gravitational factor', G', associated with the disc's constrained orbit is given by

$$G' = \frac{rv^2}{M} \cong 2.300409899 \times 10^{-13} \text{ m}^3 \text{ kg}^{-1}\text{s}^{-2}$$

According to POAMS, this disc, if allowed, would follow its natural circular orbit with radius r_0, at orbital speed v_0, relative to Earth's centre. Using equation (10) of Lecture 16 [$v_0 = GmM/L$], we obtain

$$v_0 = \frac{GmM}{L} \cong 1.349451586 \times 10^5 \text{ m s}^{-1}$$

Then, by equation (11) of Lecture 16 [$r_0 = L/mv_0$], it follows that

$$r_0 = \frac{L}{mv_0} \cong 2.198331838 \times 10^4 \text{ m}$$

Notice that these parameters are independent of the mass *m*. Hence these parameters v_0 and r_0 for the natural circular orbit of the disc are the same

as the parameters for the natural circular orbit of the 1 kg mass given in Example Two, of Lecture 16. The orbital kinetic energy, K_o, of the disc, if allowed to follow its natural circular orbit is then

$$K_o = \frac{mv_0^2}{2} \cong 9.105097915m \times 10^9 \text{ kg m}^2 \text{ s}^{-2}.$$

Suppose that now, as in the Hayasaka experiments, the disc spins at 18,000 revolutions per minute. The disc's moment of inertia, I, is given by equation (5) of Lecture 18 [$I = mr^2/2$], so that

$$I = \frac{m\rho^2}{2} \cong 5.53978898m \times 10^{-4} \text{ kg m}^2$$

Its spin kinetic energy, K_s, is then given by equation (8) of Lecture 18 [$K_s = I\omega^2/2$], so that

$$K_s = \frac{I\omega^2}{2} \cong \frac{1}{2}\left(\frac{18000}{60} \times 2\pi\right)^2 \times I \cong 9.841594603m \times 10^2 \text{ kg m}^2\text{s}^{-2}$$

POAMS predicts that the disc's spin will have an effect on its natural circular orbit, so this example now proceeds along the same lines as Example One of the previous lecture. Suppose, first of all, that the disc spins in the same direction and in the same plane as its orbit around Earths' centre-of-mass. It then follows by equation (15) of Lecture 18 [$v_s^2 = 2\mathcal{K}/m$] and equation (3) of Lecture 19 [$\mathcal{K} = K_o + K_s$] that that the disc's orbital speed, v_s, in its natural circular orbit, if allowed to follow that orbit, is given by

$$v_s^2 = \frac{2}{m}\mathcal{K} = \frac{2}{m}(K_o + K_s) \Rightarrow v_s \cong 1.349451659 \times 10^5 \text{ m s}^{-1}$$

The radius, r_s, of that natural orbit, adjusted for spin, for constant angular momentum L, is then, by equation (16) of Lecture 18 [$r_s = L/mv_s$],

$$r_s = \frac{L}{mv_s} \cong 2.198331720 \times 10^4 \text{ m}$$

Since $r_s < r_0$, it follows that the spinning disc, if allowed, would naturally orbit at a smaller distance from Earth's centre-of-mass than if it were not spinning. This is the same conclusion reached in both Section 2 and Example One of Lecture 19. In other words, in its constrained orbit, the spinning disc is more 'attracted' to Earth's surface than if it were not spinning and so *weighs more*.

Just how much more the disc weighs when it is spinning compared to when it is not can be calculated as follows. It follows by equations (11) and (12) of Lecture 16 [$r_0 = \frac{L}{mv_0}$, $v_0^2 = \frac{GM}{r_0}$] and equations (16) and (17) of Lecture 18 [$r_s = \frac{L}{mv_s}$, $\mathcal{G} = \frac{r_s v_s^2}{M}$], that the 'gravitational factor', $\mathcal{G}$, for the disc's natural circular orbit when spinning, is given by

$$\mathcal{G} = \frac{r_s v_s^2}{M} = \frac{L v_s}{mM} = \frac{r_0 v_0 v_s}{M} = \frac{v_s}{v_0} G \qquad (1)$$

In this case, with our calculated values of v_0 and v_s, equation (1) gives

$$\mathcal{G} \cong 1.000000054G.$$

The weight of the disc, when not spinning is, of course, m kg. Using equation (6) of Lecture 16 [$F = |G - G'|mM/r^2$] and the technique of Example Two in that lecture, this is the weight-equivalent of the force with magnitude F, where,

$$F = \frac{mM(G - G')}{r^2} = 9.80665m \text{ N} = m \text{ kg}$$

On the other hand, the weight of the spinning disc is the weight equivalent of the force with magnitude $\mathcal{F}$, where,

$$\mathcal{F} = \frac{mM}{r^2}(\mathcal{G} - G') = \frac{mM}{r^2}(1.000000054G - G') \cong 1.000000054F$$

$$\Rightarrow \mathcal{F} \cong 1.000000054m \text{ kg}$$

This very small increase in weight amounts to approximately 0.054 milligrams per kilogram. In the Hayasaka experiments, the mass of the disc was 0.175 grams and so the weight increase due to spin is predicted by POAMS, in this case, to be approximately one hundredth of a milligram.

Suppose now that the disc spins in the direction opposite to that of its constrained orbit about Earth's centre-of mass, and in the same plane. This time, it then follows by equation (15) of Lecture 18 and equation (4) of Lecture 19 [$\mathcal{K} = |K_o - K_s|$] that that the disc's orbital speed, v_s, in its natural circular orbit, if allowed to follow that orbit, is given by

$$v_s^2 = \frac{2}{m}\mathcal{K} = \frac{2}{m}(K_o - K_s) \Rightarrow v_s \cong 1.349451513 \times 10^5 \text{ m s}^{-1}$$

The radius, r_s, of that natural orbit, adjusted for spin, for constant angular momentum L, is then, by equation (16) of Lecture 18,

$$r_s = \frac{L}{mv_s} \cong 2.198331957 \times 10^4 \text{ m}$$

In this case, the disc is 'less attracted' to Earth, in its constrained orbit, than when it is not spinning, so that the spinning disc weighs less. This time, equation (1) gives

$$\mathcal{G} \cong 0.999999946G$$

and the weight of the spinning disc is the weight equivalent of the force with magnitude

$$\mathcal{F} \cong 0.999999946m \text{ kg}$$

Hence, the weight decrease due to spin in the Hayasaka experiments is predicted by POAMS to be approximately one hundredth of a milligram.

Before laboratory-based experiments, as outlined in Lecture Two, were first performed very recently, it was thought that is much more practical to record the effect of spin on what is conventionally perceived as the 'acceleration due to gravity' of a falling spinning axially symmetric body,

such as the disc employed in the Hayasaka experiments. As explained in Example Two of Lecture 16, as far as POAMS is concerned, this acceleration of the freely-falling body is due to the measurable reaction-force which ultimately prevents a body from following its natural force-free orbit way below Earth's surface.

For a circular disc of the same physical dimensions as the one considered in Example Two, it follows from our calculations that if the disc is not spun and is dropped towards the surface of Earth at the equator, then its acceleration is F/m and so is approximately 9.806652644 m s^{-2}. On the other hand, if it is spun in the same direction and in the same plane as Earth's equatorial rotation and is then dropped, its acceleration, now $\mathcal{F}/m$, is increased by a factor of approximately 1.000000054. Similarly, if the disc is spun in the opposite direction and in the same plane as Erath's equatorial rotation and is then dropped, its acceleration is decreased by a factor of approximately 0.999999946.

In these calculations, we have assumed that the spinning disc is situated at the equator for maximum effectiveness. In this case, the constrained orbit of the disc lies in a plane relative to Earth's centre-of-mass. It can be shown that if the disc was situated at latitude θ° north of the equator, then the magnitude of its angular momentum relative to the Earth's centre-of-mass is given by $L' = L\cos\theta$, where L is the figure calculated in this example.

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3. <u>The Hayasaka and related experiments</u>

The actual Hayasaka experiments, performed in 1989, were carried out at Sendai, which has the latitude of approximately 38° north of the equator. Replacing L by L' and performing the same sequence of calculations as above, shows that in this case, the weight of the spinning disc varies between about one two-hundredth of a milligram more or less than if it were not spinning, so the effect of spin, in the one way or the other, is predictably smaller than when the disc is situated on the equator.

In the Hayasaka experiments, the circular disc was dropped from a fixed height and it was claimed that a difference in acceleration was detected when the disc was spun in a particular direction but not in the opposite direction. The findings were published in *Physical Review Letters*. As far

as we know, no account was taken of the vector-orientations of those spins with respect to Earth's rotation. Also, the difference in acceleration claimed was very much larger than POAMS predicts. Nevertheless, these practical experiments did suggest that there *is* a change in weight of an object if it is spun.

The fact that the POAMS predicted changes in weight for Hayasaka-type experiments are extremely small accords with the findings of T. Quinn and A. Picard. Their results were published in an article in *Nature* in 1990. Jim Faller and his team at NIST (The National Institute for Standards and Technology) also claimed that the Hayasaka results were spurious in an article in the *New Scientist*, published in the same year. Hayasaka later took issue with this criticism. Viv and I always took the view that in order to properly test the predictions made by POAMS using such experiments, what is required is a set of consistent experiments involving the dropping, near the equator, of axially symmetric spinning objects which have a larger mass than Hayasaka's spinning disc. These experiments should take full account of the orientations of the spin in relation to Earth's rotation. However, as mentioned in Lecture Two, such proposals have now been superseded by lab-based experiments which have confirmed weight change due to spin.

POAMS is far from alone in predicting an effect on weight due to spin. For instance, General Relativity predicts the spinning bodies will not fall with the same acceleration as non-spinning ones. Peter Kümmel has also claimed to have measured predicted changes in weight in the form of 'gravitational acceleration' for spinning bodies. In his 2001 investigations, spins imparted to freely falling bodies produced anomalous 'gravitational' effects. Kummel's experiments involved the Fall Tower in Bremen, where the vertical is inclined at approximately 53° north of the equator. Others, such as Phillip Kanarev have argued that conservation of angular momentum changes the 'gravitational force' for rotating gyroscopes and hence their free-fall acceleration.

The departure of POAMS from Newtonian dynamics in relation to the effect of spin may at first sight appear unnecessarily radical for physics at the macroscopic level. The effect of spin on the motion of macroscopic bodies is relatively small and up until recently, as we have seen, the

experimental evidence for any such effects was somewhat scanty and controversial. However, as mentioned in the historical background Lecture 2, beginning in 2016, a series of laboratory-based experiments were carried out involving very sensitive weighing scales and gyroscopes which can be imparted with a significant spin using an electric motor. These experiments later developed into new experiments involving rotating Bismuth. These experiments clearly demonstrated the effect of spin on weight. The experimental work was carried out at MSFC and was written up as a NASA technical report in November 2021:

'A Study of the Pope-Osborne Angular Momentum Synthesis Theory (POAMS) Including a Mathematical Reformulation and Validation Experiment', R. H. Eskridge, M. A. Nelson, and M P. Schoenfeld, NASA/TM-20205010911, Marshall Flight Center, Huntsville, Alabama.

It is hoped that experiments such as these will lead to a practical use of the effect of spin on weight and gives rise to a revolution in terrestrial motion.

In the next lecture, I will present the POAMS approach at the quantum level with the incorporation of spin. I will show that the incorporation of spin is crucial at the microscopic level. We shall apply this approach to derive the parameters of the hydrogen atom in Example Three, as well as discussing the idea of relative 'attraction' and 'repulsion' in the hydrogen atom, in terms of angular momentum.

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